

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: B. Tech****Branch: Petrochemical Engineering****Semester: VI****Subject Code & Name: BTPCPE605A_Y22 Energy Management in Petrochemical Industry****Max Marks: 60****Date: 30/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Minor actions such as monitoring monthly energy bills and upgrading to energy-saving lightbulbs are includes in _____. a. Energy Conservation b. Energy Management c. Energy Services d. Can't say	(Understanding/CO1)	1
2	India is the _____largest energy consumer in the world. a. First b. Second c. fourth d. Third	(Remember/CO2)	1
3	The largest energy source is _____ followed by petroleum and traditional biomass. a. Coal b. Water c. Methane d. Propane	(Remember/CO3)	1
4	Energy management is a key component of_____ a. Environmental management b. Carbon management c. Nitrogen management d. Water management	(Apply/CO4)	1
5	The main objective of energy management is to_____. a. Minimize energy cost b. Minimum environmental effects c. Maintain optimum energy procurement and utilization d. All of these	(Understanding/CO5)	1
6	Producer gas is a mixture of_____ a. carbon monoxide and hydrogen gas b. carbon monoxide and nitrogen gas c. carbon monoxide and water vapor d. carbon monoxide and nitrous oxide	(Understanding/CO1)	1
7	_____is the main constituent of 'Natural gas' from oil and gas fields. a. Methane b. Hydrogen c. Ethane d. Propane	(Remember/CO2)	1
8	The term biomass most often refers to _____	(Remember/CO3)	1

	a. Inorganic matter	b. Organic matter	c. Chemicals	d. Ammonium compounds		
9	Which of the following is NOT Factors affecting energy conservation economics.				(Understanding/CO4)	1
	a. Energy prices	b. Technology costs	c. Energy efficiency policies and regulations	d. none of the given		
10	All types of coals can be converted into coke.				(Apply/CO5)	1
	a. True	b. False	c. Neither a or b	d. Can't say		
11	Thermal shield is used in high powered nuclear reactors to _____.				(Apply/CO1)	1
	a. absorb the fast neutrons	b. protect the walls of the reactor from radiation damage	c. slow down the secondary neutrons	d. protect the fuel element from coming in contact with the coolant		
12	State the following: Coal Gasification and liquefaction can produce fuels as equivalent to oil and natural gas.				(Apply/CO2)	1
	a. True	b. False	c. Neither a or b	d. Can't say		
Q. 2	Solve the following.					12
A)	Explain the term “Energy Management” with its principles.				Understanding /CO1	6
B)	Give the note on: Energy conservation opportunities by sector.				Remember /CO1	6
Q.3	Solve the following.					12
A)	Describe the Hydrogen Important as A Future Clean Energy				Remember /CO2	6
B)	Describe “Energy Audit” with its need in terms of Energy Management.				Apply /CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Define Life Cycle Costing (LCC) with its importance in basic discounting of Energy conservation methods.				Remember /CO3	6
B)	Describe the factors affecting on energy conservation economics. (Any 6)				Understanding /CO3	6
C)	Explain Energy Conservation Costing Methods associate with implementing energy-saving measures.				Apply /CO3	6

Q.5	Solve Any Two of the following.		12
A)	Enlist key consideration of Energy Conservation of Available Work in The Plants.	Remember /CO4	6
B)	Describe general meaning of Energy Pricing with its purpose in Energy Management.	Understanding /CO4	6
C)	Enlist processes involving recognizing of energy losses or inefficiencies that cannot be fully recovered or reversed as indications of irreversible processes in energy conservation. (Any 6)	Apply /CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Describe in detail about “Coal” as primary source of energy with its based techniques used for energy generation.	Understanding /CO5	6
B)	Examine the importance of “Thermal Storage” device. Describe operation of <i>Babcock</i> thermal storage boiler.	Remember /CO5	6
C)	Explain the term “Future of Natural Gas” as source of energy.	Apply /CO5	6
	*** End ***		